Aastha

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B.Tech, Computer Science Engineering (2021-2025)

EDUCATION			
B.Tech	2025	Chandigarh University	8.1 CGPA
Class XII (CBSE)	2021	Adarsh Vidya Mandir, Bhiwani	94.0%
Class X (CBSE)	2019	Vaish Model Senior Secondary School, Bhiwani	92.8%

SKILLS

1. **Python**
2. **Pytorch**
3. **Deep Learning**
4. **OpenCv**
5. **Machine Learning**
6. **Data Analysis**
7. **MySQL**
8. **Power BI/Tableau**

INTERNSHIPS

PAUL MERCHANTS, **(**Chandigarh), Jul'25 - Present

1. **Loan Prediction Project:**
   Built a machine learning model to predict loan approval using customer demographic and financial data, improving decision-making efficiency.
2. **PaulBot Project:**
   Developed a domain-specific chatbot using FastAPI, Groq LLM, and MongoDB to assist users with queries about Paul Merchants, with contextual memory and real-time conversation summarization.

TVS MOTOR, *Hosur(Tamil Nadu), Aug 2024- Dec 2024*

1. **Enhanced a real-time engine piston circlip detection system** using **YOLO-based object detection**, OpenCV, and Tkinter/PyQt5, ensuring accurate classification of circlip presence.
2. **Integrated live camera feed processing with a GUI interface** for real-time monitoring, database storage (SQL Server), and WebSocket-based API communication for automated result updates.
3. **Optimized detection pipeline for efficiency and stability**, reducing frame rate bottlenecks, handling asynchronous WebSocket tasks, and implementing multi-camera support for enhanced inspection accuracy.

PROJECTS

1. **Parkinson's Disease Prediction (Machine Learning)**
   - Developed a **Parkinson's Disease Prediction** model using Machine Learning, leveraging feature extraction and classification techniques to identify early symptoms.
   - Implemented and optimized algorithms for accurate detection, improving predictive performance through data preprocessing and model tuning.
2. **SmartCV Builder (Frontend Development)**
   - Fortified SmartCV Builder with a drag-and-drop module and customizable template gallery, increasing user satisfaction by 40% based on post-interaction surveys and user testing sessions.
   - Implemented an interactive UI using modern web technologies to enhance user experience and customization options.
3. **Phishing Detection using Machine Learning**
   - Created a Machine Learning-based Phishing Detection System to identify fraudulent websites using advanced classification algorithms.
   - Implemented **feature extraction and model training** techniques to enhance detection accuracy, improving cybersecurity measures.

POSITIONS OF RESPONSIBILITY

1. **Event Coordinator, IEEE Photonics Society, Chandigarh University**
   - Led the planning and execution of technical workshops, seminars, and networking events, fostering industry-academia collaboration.
   - Coordinated photonics and optics-related technical sessions, enhancing knowledge-sharing among students and professionals.
   - Managed event logistics, speaker coordination, and audience engagement, ensuring successful knowledge dissemination.
2. **Technology Lead, Scriptogen (Research and Development Society), Chandigarh University**
   - Led a team of researchers and developers in executing innovative R&D projects across various technology domains.
   - Spearheaded initiatives in AI, IoT, and software development, mentoring students in hands-on projects and research papers.
   - Organized technical bootcamps and hackathons, facilitating skill development and fostering a culture of innovation.

HONORS AND ACADEMIC ACHIEVEMENTS

1. Conceived a novel space debris removal system leveraging AI-powered drone swarms, earning 1st place in NASA's International Space Challenge amongst 5,000+ participants globally and besting 500 competing teams worldwide.
2. Conceived and executed a novel encryption method to safeguard student data for an AI education platform, which the judges remarked was the most innovative solution for data security at Hackoverflow.
3. Pioneered a unique data encryption method using DNA sequencing at Hack-2-Skills Instinct Hackathon, winning the 'Out of Box' award due to the solution's potential in secure data transmission.
4. Awarded **Gold Medal** for outstanding performance in **NPTEL's IoT course**, demonstrating expertise in Internet of Things and technical proficiency.
5. Recognized for **outstanding academic performance**, reflecting strong technical knowledge and consistent excellence.
6. Honored for exceptional contributions and leadership in **IEEE initiatives**, highlighting teamwork, dedication, and organizational skills.

PUBLICATIONS

1. Aastha, C., Mahak, and Gurmeet. "**Evaluation of Flying cars**" 2nd International Conference for " Innovation in Technology **(INOCON) 2023**".
2. Aastha . "**Signal Class Labelling with Active Learning and Classification using Machine Learning**" India Utility Patent- 202341040386 A (Published).